Linear Graphs 3: Equation from a Graph

Name _____ Date _____ No calculator Version 1

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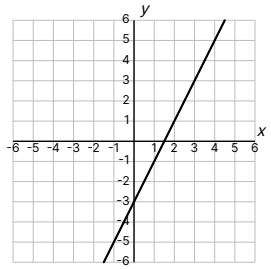
KEY WORDS gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c · intercepts *(pūtahi)*

A Gradient and intercept ●○○

Remember: Read c where the line crosses the y-axis. Find m from rise over run. Then write $y = mx + c$.

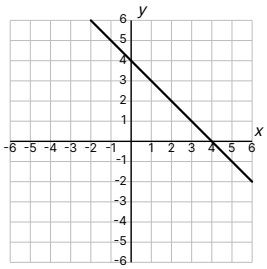
1. Write the equation of each line in the form $y = mx + c$.

a)



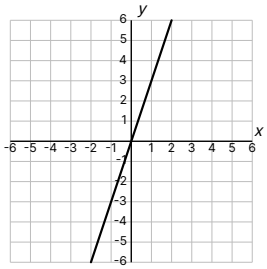
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b)



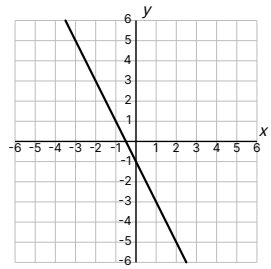
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c)



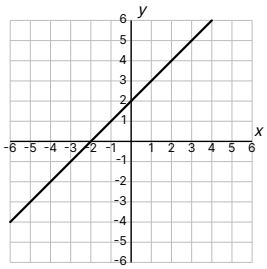
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d)



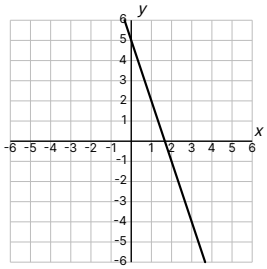
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e)



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f)



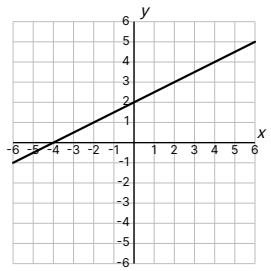
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B Fractional gradients ●●○

Remember: Count a run that lands on a grid corner. Rise 1 for run 2 gives $m = \frac{1}{2}$; down 2 for run 3 gives $m = -\frac{2}{3}$.

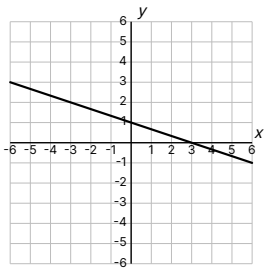
2. Write the equation of each line.

a)



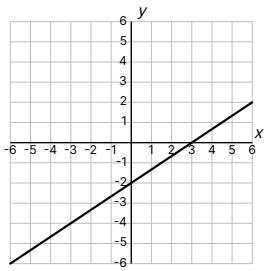
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b)



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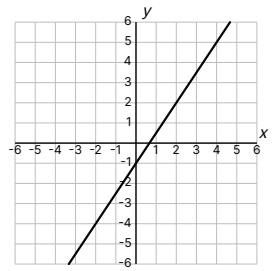
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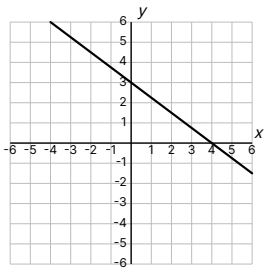
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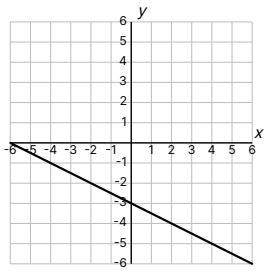
d)



e)



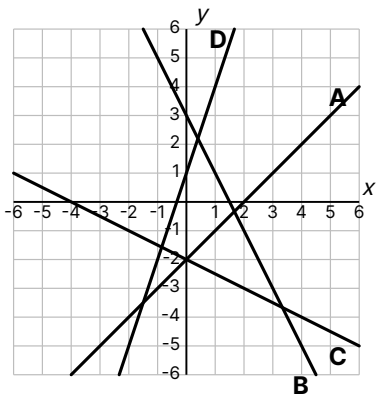
f)



C Several lines on one grid ●●○

Remember: Each line is labelled where it leaves the grid. A horizontal line is $y =$ a number; a vertical line is $x =$ a number.

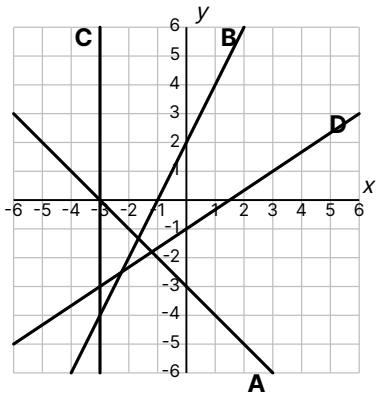
3. Write the equation of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

4. Write the equation of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

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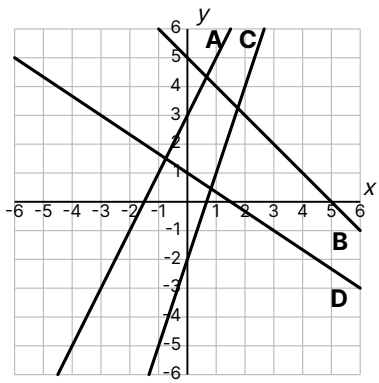
D Equations in other forms ●●●

Remember: Rearrange each equation into $y = mx + c$ first: $2y = 4x + 6$ is $y = 2x + 3$, and $x + y = 5$ is $y = -x + 5$. Then look for its gradient and intercept on the grid.

5. Each line on the grid has one of these equations: $2x + 3y = 3$, $2y = 4x + 6$, $2y = 4x - 6$, $3x - y = 2$, $x + y = 5$, $x - y = 5$. Match each line to its equation. Two of the equations are not used.

a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____



E From the two intercepts ●●●

Remember: The y-intercept gives c straight away. The gradient is the change in y over the change in x between the two intercepts: from $(a,\ 0)$ to $(0,\ b)$ it is $\frac{b}{-a}$.

6. A line crosses the axes at the two points given. Find its equation.

a) $(4,\ 0)$ and $(0,\ 2)$ **b)** $(3,\ 0)$ and $(0,\ -6)$ **c)** $(-2,\ 0)$ and $(0,\ 4)$

d) $(5,\ 0)$ and $(0,\ 5)$ **e)** $(-3,\ 0)$ and $(0,\ -6)$ **f)** $(6,\ 0)$ and $(0,\ -4)$

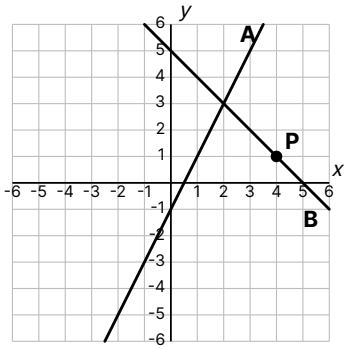
★ Extension

7. Lines A and B and the point P are drawn on the grid.

a) Write the equation of line A and of line B.

b) Write down the coordinates of the point where the lines cross, and check that it fits both equations.

c) Find the equation of the line through P that is parallel to line A.



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Linear Graphs 3: Equation from a Graph

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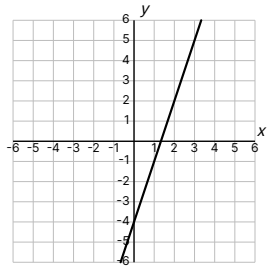
KEY WORDS gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c · intercepts *(pūtahi)*

A Gradient and intercept ●○○

Remember: Read c where the line crosses the y-axis. Find m from rise over run. Then write $y = mx + c$.

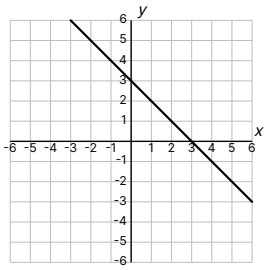
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a)



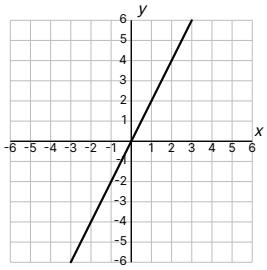
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b)



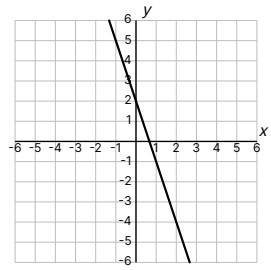
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c)



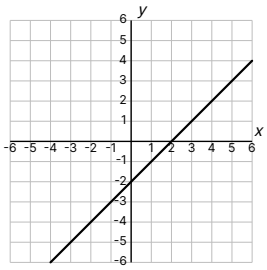
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d)



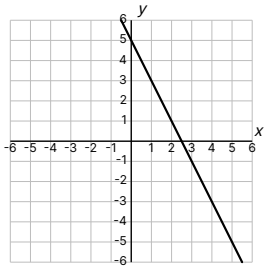
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e)



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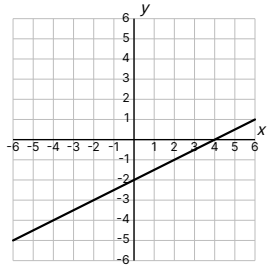
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B Fractional gradients ●●○

Remember: Count a run that lands on a grid corner. Rise 1 for run 2 gives $m = \frac{1}{2}$; down 2 for run 3 gives $m = -\frac{2}{3}$.

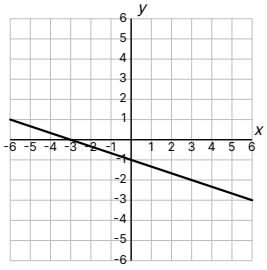
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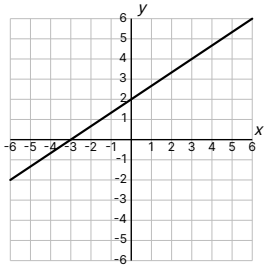
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b)



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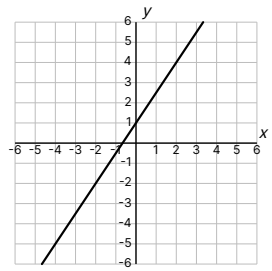
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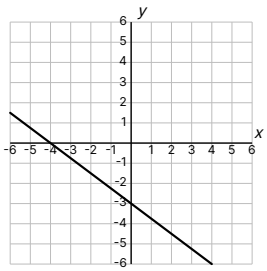
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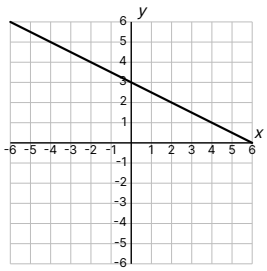
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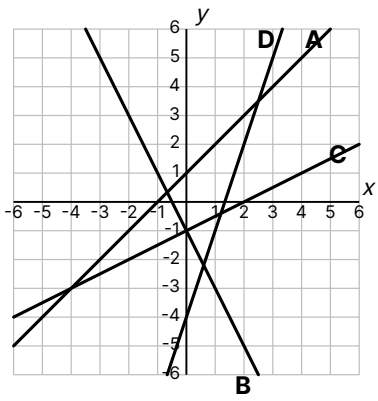
f)



C Several lines on one grid ●●○

> **Remember:** Each line is labelled where it leaves the grid. A horizontal line is $y =$ a number; a vertical line is $x =$ a number.

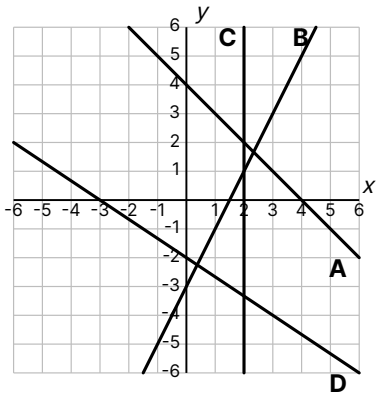
3. Write the equation of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

4. Write the equation of each line on the grid.



a) Line A _____ **b)** Line B _____

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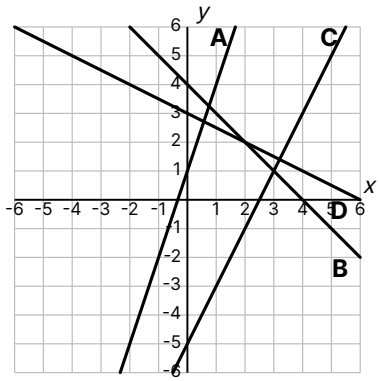
D Equations in other forms ●●●

Remember: Rearrange each equation into $y = mx + c$ first: $2y = 4x + 6$ is $y = 2x + 3$, and $x + y = 5$ is $y = -x + 5$. Then look for its gradient and intercept on the grid.

5. Each line on the grid has one of these equations: $2x - y = 5$, $3y = 9x + 3$, $3y = 9x - 3$, $x + 2y = 6$, $x + y = 4$, $x - y = 4$. Match each line to its equation. Two of the equations are not used.

a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____



E From the two intercepts ●●●

Remember: The y-intercept gives c straight away. The gradient is the change in y over the change in x between the two intercepts: from $(a,\ 0)$ to $(0,\ b)$ it is $\frac{b}{-a}$.

6. A line crosses the axes at the two points given. Find its equation.

a) $(2,\ 0)$ and $(0,\ 4)$ **b)** $(4,\ 0)$ and $(0,\ -2)$ **c)** $(-3,\ 0)$ and $(0,\ 6)$

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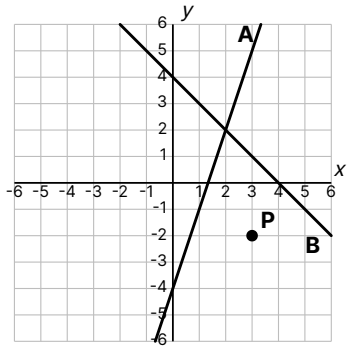
★ Extension

7. Lines A and B and the point P are drawn on the grid.

a) Write the equation of line A and of line B.

b) Write down the coordinates of the point where the lines cross, and check that it fits both equations.

c) Find the equation of the line through P that is parallel to line A.



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